

SKAGWAY, AK, USA

WMO: 703620

Lat: 59.467N

Lon: 135.300W

Elev: 6

StdP: 101.25

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 25335

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-14.0	-27.7	0.3	-15.4	-25.1	0.4	-13.1	18.0	-4.6	16.7	-3.3	10.8	30	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	22.3	14.3	20.2	13.7	18.6	13.1	15.3	20.2	14.6	18.7	13.9	17.5	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	9.4	16.2	12.5	9.0	15.6	12.0	8.7	15.1	42.6	20.2	40.5	18.7	38.6	17.5	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.4	11.9	DB	-17.2	26.9	3.0	3.0	-19.4	29.0	-21.1	30.8	-22.8	32.5	-24.9	34.7	(4)
(5)			WB	-18.2	16.8	2.8	2.4	-20.2	18.5	-21.8	19.9	-23.4	21.3	-25.4	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-2.5	-1.1	0.6	5.3	10.1	13.3	14.4	13.9	10.6	6.3	0.9	-1.5	(6)	
	DBStd	7.22	6.38	4.92	4.47	2.74	2.75	2.27	1.93	2.09	2.17	2.67	4.49	4.96	(7)	
	HDD10.0	1926	387	311	291	143	33	1	0	1	17	116	273	355	(8)	
	HDD18.3	4541	644	544	549	389	256	154	124	139	232	373	523	613	(9)	
	CDD10.0	430	0	0	0	3	36	98	136	121	36	2	0	0	(10)	
	CDD18.3	4	0	0	0	0	0	2	2	1	0	0	0	0	(11)	
	CDH23.3	43	0	0	0	0	2	15	11	16	0	0	0	0	(12)	
	CDH26.7	8	0	0	0	0	0	2	2	4	0	0	0	0	(13)	
(14) Wind	WSAvg	4.6	6.3	5.4	5.2	4.2	4.4	4.6	4.2	3.7	3.7	4.1	4.6	5.4	(14)	
(15) Precipitation	PrecAvg	979	91	77	65	50	38	34	42	67	130	160	115	103	(15)	
	PrecMax	1108	200	143	213	113	90	81	131	142	204	293	292	230	(16)	
	PrecMin	760	29	16	8	14	12	10	6	13	69	88	9	29	(17)	
	PrecStd	97	46	34	43	25	20	14	28	33	36	49	55	45	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	7.9	12.0	19.2	22.5	25.5	24.4	25.2	20.2	13.3	9.6	9.3	(19)	
		MCWB	6.4	4.7	6.1	9.7	11.9	15.0	15.6	13.8	11.9	8.8	6.8	6.8	(20)	
	2%	DB	7.2	6.8	9.0	14.5	19.5	21.8	21.3	21.2	17.1	11.9	8.2	7.2	(21)	
		MCWB	5.0	4.0	4.7	6.8	11.0	14.7	15.1	14.4	11.3	8.3	5.7	5.1	(22)	
	5%	DB	6.0	5.8	7.2	12.0	17.3	19.2	19.2	18.9	15.2	11.0	7.2	5.9	(23)	
		MCWB	3.9	3.4	3.8	5.9	10.5	13.4	14.3	13.7	11.1	8.0	5.0	4.0	(24)	
	10%	DB	4.7	4.6	5.9	9.8	15.2	17.3	17.7	17.4	13.9	10.0	6.2	4.7	(25)	
		MCWB	2.9	2.6	3.2	5.5	9.7	12.5	13.6	13.3	10.8	7.3	4.3	3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	5.6	7.4	10.1	13.2	16.8	16.5	15.9	13.9	10.6	7.6	7.0	(27)	
		MCDB	9.1	7.4	10.8	18.3	19.0	21.7	22.1	21.9	17.0	12.1	9.1	8.8	(28)	
	2%	WB	5.2	4.7	5.5	7.9	12.0	14.9	15.4	15.0	12.7	9.6	6.4	5.4	(29)	
		MCDB	7.0	6.3	8.1	12.8	18.1	20.6	20.0	19.4	15.4	11.1	7.7	7.2	(30)	
	5%	WB	4.1	3.8	4.4	6.8	10.9	13.8	14.6	14.3	11.9	8.7	5.4	4.1	(31)	
		MCDB	5.8	5.3	6.7	10.6	15.9	18.6	18.6	17.9	14.1	10.3	6.8	5.6	(32)	
	10%	WB	3.1	2.9	3.5	5.9	10.2	12.8	13.9	13.7	11.2	7.8	4.6	3.1	(33)	
		MCDB	4.4	4.3	5.5	9.2	14.6	16.8	17.3	16.8	13.3	9.5	6.0	4.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.8	5.5	7.2	7.6	6.8	5.8	6.0	5.3	4.7	3.9	4.0	(35)	
		MCDBR	5.6	5.4	7.9	12.0	12.5	11.8	9.9	10.1	8.3	5.8	4.4	5.3	(36)	
	5% WB	MCWBR	4.1	3.7	4.7	6.1	5.9	5.7	4.6	4.6	4.0	3.3	3.1	4.0	(37)	
		MCWBR	5.2	4.9	6.6	9.4	10.3	10.4	8.7	8.3	6.2	4.6	4.2	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.300	0.332	0.374	0.386	0.423	0.413	0.392	0.350	0.312	0.272	0.253	(40)	
		taud	1.978	2.078	2.013	1.947	2.045	2.068	2.173	2.248	2.315	2.187	2.009	1.943	(41)	
	Edn at Noon		449	684	778	811	835	806	806	795	770	665	465	347	(42)	
		Edh at Noon	53	82	119	152	149	149	131	113	89	72	49	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.39	1.14	2.47	4.12	5.49	5.21	4.71	3.59	2.31	1.09	0.50	0.24	(44)	
	RadStd		0.03	0.16	0.26	0.40	0.47	0.33	0.36	0.36	0.23	0.13	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	+2.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	+1.26	+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page